

■ Asset Management System

Frontend Requirements Blueprint

1. Pages / Screens Ki List (Jo Frontend Mein Honi Chahiye)

- **Dashboard / Assets List Page:** Saare assets ki table, pagination (Current Page, Total Pages), search bar, aur category/status filters ke sath.
- **Add / Update Asset Page / Modal:** Naya asset banane ya purane asset ko edit karne ke liye form.
- **Asset Allocation Page:** Assets ko employees ko assign karne aur unke return/history track karne ke liye.
- **Maintenance & Bills Page:** Kisi asset par hone wale kharche (repair) aur purchase bills/invoices ko manage karne ke liye.
- **Categories & Types Management:** Nayi Categories aur Types add karne ke liye dropdowns/forms.

2. Forms & Fields Breakdown (Kya-Kya Input Lena Hai)

A. Add / Update Asset Form (assetController.js)

- AssetTag — Text, **Required** — e.g., LAP-1001
- AssetName — Text, **Required** — e.g., MacBook Pro
- TypeID — Dropdown / Number, **Required** — Asset Types se aayega
- VendorName — Text, Optional
- PurchaseDate — Date Picker, Optional
- Price — Decimal/Number, Optional
- WarrantyExpiryDate — Date Picker, Optional
- Status — Dropdown: Available, Assigned, Decommissioned, etc.
- **Dynamic Attributes (optional based on type):** Electronics: RAM, Processor, Storage, MAC_Address; Communication: PhoneNumber, ServiceProvider, IMEI_Number; Furniture/Others: Material, Color, Dimensions

B. Asset Allocation Form (allocationController.js)

- assetID — Dropdown of Available assets
- employeeID — Dropdown of Employees from /employees API
- assignedDate — Date Picker, optional — defaults to today
- remarks — Textarea, optional

C. Add Maintenance Form (maintenanceController.js)

- assetID — Dropdown of Assets
- cost — Number, **Required**
- description — Textarea
- maintenanceDate — Date Picker
- receipt — File upload for receipt/bill PDF or image

D. Add Bill / Invoice Form (billController.js)

- assetID — Dropdown of Assets
- billNumber — Text, **Required**
- billDate — Date Picker
- amount — Decimal/Number
- invoiceFilePath — File upload for invoice document

E. Category & Type Forms (categoryController.js & typeController.js)

- **Category Form:** categoryName — Text, unique check
- **Type Form:** typeName — Text + categoryID — Dropdown of categories

3. API Endpoints Map (Frontend se kahan request marni hai)

Feature	Method	Endpoint URL	Request Body / Query Params
Get All Assets	GET	/api/assets	?page=1&limit;=10&search;=&category;=&status;=
Add Asset	POST	/api/assets	JSON Object (AssetTag, AssetName, TypeID, etc.)
Update Asset	PUT	/api/assets/:id	JSON Object (Updated fields)
Delete (Soft)	DELETE	/api/assets/:id	-
Get Categories	GET	/api/categories	-
Add Category	POST	/api/categories	{ categoryName }
Get Types	GET	/api/types	?categoryId=1 (Optional)
Add Type	POST	/api/types	{ typeName, categoryId }
Get Employees	GET	/api/employees	-
Allocate Asset	POST	/api/allocations	{ assetID, employeeID, assignedDate, remarks }
Return Asset	PUT	/api/allocations/return/:allocationId	-
Asset History	GET	/api/allocations/history/:assetId	-
Add Maintenance	POST	/api/maintenance	FormData (assetID, cost, description, file)
Add Bill	POST	/api/bills	FormData (assetID, billNumber, amount, file)

Integration Note: Frontend should use the endpoint paths, methods, fields and request formats above as the integration contract. Keep the API base URL configurable so it can be changed when the backend moves to another machine/server.